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Middle Triassic paleomagnetic results from central Hubei province, China and their tectonic implications

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Abstract. Twenty one sites consisting of 4 to 7 samples per site were drilled from the redbeds of the Middle Triassic Badong Formation from central Hubei province, China. These samples were subjected to progressive thermal demagnetization which revealed two components of magnetization. The low temperature (A) component was probably acquired at a late stage of folding and is similar in direction to the late Mesozoic overprint widely observed in the eastern Yangtze Block . The high temperature (B) component is prefolding with dual polarity. Unlike the prefolding paleomagnetic directions observed in eastern Sichuan, western Hubei and northwestern Hunan, the B component resolved from central Hubei indicates no significant rotation with respect to other parts of the Yangtze Block. Since one of the two sampled sections is in the strongly deformed foreland fold-thrust belt of the Qinling Fold Belt , this would seem to negate the possibility that the differential rotations observed in the border area between Sichuan, Hubei, Hunan and Guizhou are associated with the collision of the Yangtze Block with the North China Block.

Introduction

We have reported paleomagnetic results from the Middle Triassic rocks from western and southeastern Hubei province and northwestern Hunan province in the Yangtze Block (YB) (Huang and Opdyke, 1996). The results indicate that western Hubei and northwestern Hunan have been subjected to significant differential rotations but that southeastern Hubei may not have been significantly rotated within paleomagnetic error limits since the acquisition of the remanence. The differential rotations observed in eastern Sichuan and the border area between Hubei, Hunan, Guizhou and Sichuan have been inferred to be due to oroclinal bending in the late Mesozoic (Steiner et al., 1989; Huang and Opdyke, 1996). To further constrain the extent and cause of such rotations, we studied coeval rocks from central Hubei province.

Sampling

The samples were collected from the Middle Triassic Badong Formation in the Jingmen-Dangyang basin which is separated by the domelike Huangling anticline from the Zigui basin, at the west flank of which paleomagnetic results have been derived from the type section for the Badong Fm (Huang and Opdyke, 1996) (Fig. 1). The Jingmen-Dangyang basin is a NNW-trending synclinal basin with Jurassic rocks in the core and Triassic rocks cropping out along the flanks. The rocks in the core of the syncline dip very gently; the rocks on the northern flank, however, are involved in the deformation of the foreland fold-thrust belt at the northern margin of the YB and constitute a bundle of WNW-trending elongate folds. The deformation is further intensified as approaching the Fangxian-Guangji fault to the north, a north-dipping boundary thrust between the YB and the Qinling Fold Belt (QFB), and the WNW-trending folds become overturned with southward vergence (Hubei Bureau of Geology and Mineral Resources, 1990). Clearly, these folds own their origin to the collision of the YB with the North China Block (NCB) and subsequent overthrusting of the QFB onto the YB (Mattauer et al., 1985; Hsu et al., 1987; Liu and Hao; 1989; Okay et al., 1993). Superimposed on the west flank of the Jingmen-Dangyang basin is the Yuan'an graben, a NNW-trending fault depression filled with Upper Cretaceous molassic sediments. The Upper Cretaceous unconformably rests on the Middle Jurassic-Triassic rocks in the Jingmen-Dangyang basin (Hubei Bureau of Geology and Mineral Resources, 1990; Zhang et al., 1987). Therefore, folding of the Jura-Triassic rocks in the Jingmen-Dangyang basin took place sometime in the interval from late Jurassic to early Cretaceous (Yanshanian).

The Badong Fm in the Jingmen-Dangyang basin is correlated with the type section in the Zigui basin in terms of lithology and biota (Hubei Bureau of Geology and Mineral

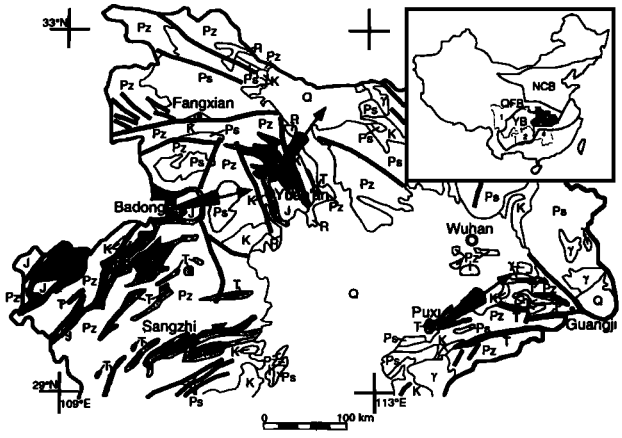


Figure 1. Sketch geological map of Hubei and northern Hunan provinces. Solid circles denote the sampled sections of this study. Strata: Ps = pre-Sinian; Pz = Sinian- Permian; T = Triassic; J = Jurassic; K = Cretaceous; R = Tertiary; Q = Quaternary. Triassic outcrops are shaded. Granitic intrusions are indicated with γ. Heavy lines are major faults. The mean paleomagnetic declinations and associated uncertainties observed from the Middle Triassic rocks in the area are indicated with arrows. The shaded area in the inset is the area the map shows. Gray lines are the boundaries of Sichuan (1), Guizhou (2), Hubei (3) and Hunan (4) provinces. Abbreviations for the tectonic units: NCB = North China Block; YB = Yangtze Block; QFB = Qinling Fold Belt.

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Resources, 1990; Zhang et al., 1987). It is divisible into five members with Member 3 being characterized by limestone and Members 2 and 4 by purplish red siltstone and mudstone. The thickness of the Badong Fm at the sampled sections of this study varies from 1087 to 1241 m. We collected paleomagnetic samples, using a gasoline-powered rock drill and an orientation device mounted with a magnetic compass, from Member 2 at two sections: the Maoping section at the west flank of the basin about 10 km NNE of downtown Yuan'an (county) and the Liujiatai section (31.4°N, 111.7°E) at the north flank about 26 km further NNE of the Maoping section. Eleven sites were drilled from the fresh exposures in a stream bed at the Maoping section where bedding attitudes vary from dipping southwest to northeast at 13-18°, and ten more sites were taken from the roadcut at the Liujiatai section where the rocks dip south at 36-46°. Six samples were usually taken from a site and a total of 125 samples were collected.

Results

All the samples were subjected to progressive thermal demagnetization. Standard analytical techniques (Zijderveld, 1967; Kirschvink, 1980; Fisher, 1953) were employed to analyze the data.

Except for the marly samples from sites 1-3 taken from right below Member 3 at the Maoping section, which are relatively weakly magnetized with initial intensities of natural remanent magnetization (NRM) in the range of 1.0-2.6 mA/m, the samples from other sites have initial NRM intensities ranging from 2.2 to 7.8 mA/m. Like the samples from the type section at Badong in the Zigui basin (Huang and Opdyke, 1996), most of the samples in this study possess two components: a northeasterly and steeply downward (in-situ coordinates) component, designated as component A hereafter, was removed in the temperature range 100-650°C; above

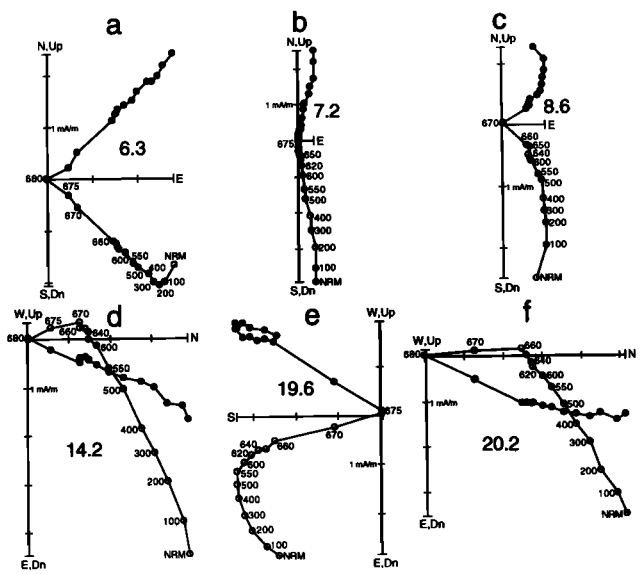


Figure 2. Representative orthogonal plots of thermal demagnetization (in-situ coordinates). a-c are from the Maoping section and d-f are from the Liujiatai section. Solid (open) circles are plotted on horizontal (vertical) planes. Treatment levels are in degrees Celsius and indicated along the demagnetization trajectory. The large-sized numbers (e.g. 6.3) are the idenfication numbers of the samples shown.

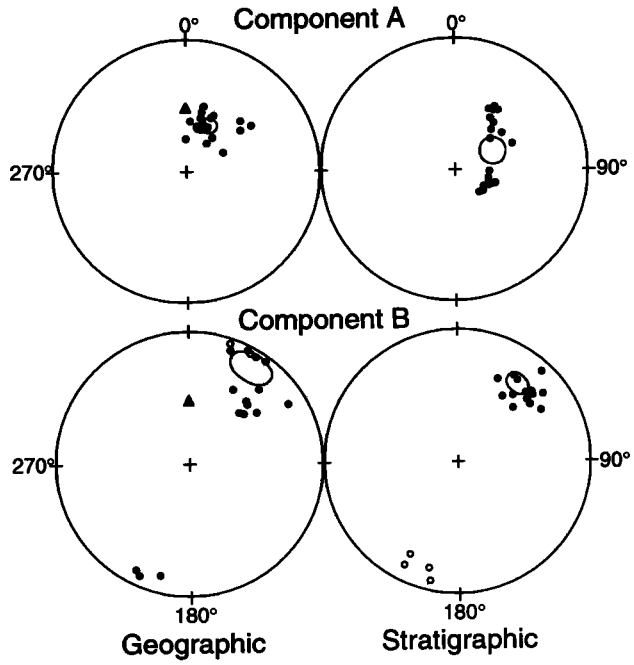


Figure 3. Equal-area projections of site mean directions of components A and B. Solid (open) circles represent directions plotted onto the lower (upper) hemisphere. Black triangle denotes the present dipole field direction at the sampled sections. The overall 95% confidence circles are also shown.

650°C, a characteristic remanent magnetization (ChRM) component, designated as component B, was resolved (Figs. 2c, d, e and f). Component A often accounts for 30-80% of the initial intensities (Figs. 2c, d, e and f); it is entirely dominant in some samples (Fig. 2b); sample 6.3 is the only sample from the collection which basically exhibited component B after removal of a spurious magnetization at 100°C (Fig. 2a). Two polarities are present in the B component. Most of the sites are of normal polarity with northeasterly and moderately downward in-situ directions (for the Maoping section) or north-northeasterly and shallow directions (for the Liujiatai section) (Figs. 2a, c, d and f). The samples from the four sites (16-19) taken from a 20 m-thick interval at the Liujiatai section are all of reversed polarity with south-southwesterly and shallow directions in in-situ coordinates (Fig. 2e). After correction for bedding tilt, the mean directions for the B component from the two sections converge (Fig. 3 lower) with a $\kappa s/\kappa g$ ratio of 2.75 which is greater than the threshold value (2.26) for a positive fold test at 99% confidence level (McElhinny, 1964). By contrast, the site mean directions for the A component are split into two groups dictated by the fold flanks with $\kappa s/\kappa g$ = 0.32 (Fig. 3 upper). Therefore, component B was probably acquired before folding and component A was acquired after folding. The tilt-corrected global mean direction for the B component is D/I = 38.5°/26.0° which yields a pole at 50.5°N/218.8°E, and the in-situ global mean direction for the A component is D/I = 27.5°/58.3° which corresponds to a pole at 66.6°N/173.0°E. Site and formation statistics are given in Table 1.

Discussion

Incremental unfolding has yielded a maximum value for the precision parameter κ (60.4) and a global mean direction of D/I

Table 1. Site and Formation Statistics for Badong Fm From Central Hubei (31.4°N,111.7°E)

Site	Bedding		n/N	In Situ D(°)	In Situ I(°)	Tilt Corrected D(°)	Tilt Corrected I(°)	α_{95}	κ_2	Pole Position Lat.(°N)	Pole Position Long.(°E)
1	152/13	A	6/6	47.4	44.1	42.8	36.5	6.3	113.2	48.3	196.7
2	150/13	B	2/5	58.8	15.0	58.7	28.0			34.2	204.5
		A	5/5	54.7	40.3	53.3	53.3	5.3	210.0	41.1	197.3
3	153/13	A	7/7	53.6	48.0	50.1	60.8	3.8	247.9	44.0	189.9
4	297/18	A	6/7	38.0	63.6	34.0	45.8	10.0	45.9	57.4	165.3
5	317/17	B	4/6	44.6	45.4	45.1	28.3	8.4	121.0	45.9	212.5
		A	6/6	36.2	69.0	40.7	52.2	4.5	226.5	56.2	152.0
6	317/17	B	6/7	52.7	36.7	51.8	19.7	8.0	71.3	37.7	213.7
		A	5/7	26.0	59.6	32.6	43.4	12.1	41.0	67.1	170.7
7	319/16	B	5/6	43.3	24.5	43.8	8.6	22.9	12.1	40.9	225.8
		A	6/6	20.9	61.7	29.9	47.0	5.9	129.3	69.7	160.7
8	345/14	B	6/6	44.2	38.0	48.4	25.7	10.7	39.8	42.3	212.2
		A	6/6	19.7	60.9	35.2	51.4	9.2	53.4	71.0	161.9
9	345/14	B	3/6	47.9	44.0	52.4	31.2	10.0	152.4	40.5	205.8
		A	6/6	62.6	64.6	66.6	50.8	6.1	121.6	40.3	165.0
10	345/14	B	6/6	31.0	35.4	36.4	24.9	15.7	19.0	52.0	221.9
		A	6/6	21.3	52.9	33.0	43.5	4.8	199.2	71.9	188.9
11	345/14	B	4/6	42.4	36.8	46.6	24.7	16.9	30.4	43.5	214.1
		A	5/6	25.3	53.9	36.5	43.8	8.4	83.8	68.6	186.2
12	73/42	B	6/6	20.1	9.7	34.6	40.6	8.9	58.2	58.3	208.0
		A	6/6	15.5	56.1	107.4	68.7	2.9	529.3	76.1	175.2
13	88/46	B	6/6	32.2	6.3	46.6	41.9	9.5	50.4	48.4	199.7
		A	6/6	26.8	51.8	120.2	69.3	2.8	572.2	67.3	192.0
14	79/40	B	6/6	29.3	-6.3	33.6	23.9	10.7	40.0	53.8	225.6
		A	6/6	20.7	62.9	128.1	68.6	3.7	322.0	69.3	157.2
15	79/42	B	6/6	36.9	5.4	48.6	31.1	8.9	57.3	43.7	208.3
		A	6/6	15.8	62.8	133.8	69.1	4.1	272.7	72.2	151.3
16	55/36	B	6/6	195.1	14.7	193.7	-9.1	21.2	10.9	-60.3	83.3
		A	6/6	359.6	70.5	115.2	67.5	9.6	49.9	66.7	111.1
17	74/41	B	6/6	204.3	8.9	207.7	-22.2	12.1	31.7	-57.6	53.4
		A	6/6	20.8	59.0	113.2	66.6	8.0	71.5	71.2	169.5
18	67/41	B	4/4	195.3	14.5	196.1	-18.1	10.2	81.8	-63.3	74.2
		A	4/4	15.0	61.5	111.8	65.5	4.2	482.6	73.6	154.2
19	60/46	B	6/6	207.2	12.4	207.6	-13.4	5.8	135.7	-54.4	59.6
		A	6/6	5.5	59.4	108.2	63.6	7.3	84.1	80.1	137.0
20	75/45	B	5/6	19.3	-4.7	26.3	31.6	12.1	40.7	62.2	226.5
		A	5/6	15.0	52.1	105.8	69.0	10.1	58.3	77.2	191.8
21	75/45	B	5/6	27.2	4.4	40.1	35.2	6.9	122.6	52.1	210.3
		A	5/6	16.3	47.7	93.3	68.4	3.2	581.7	75.7	208.0
Fm Mean		B	18/21	33.1	13.6	38.5	26.0			50.5	218.8
				κ_3 = 10.8		κ_3 = 29.8		α_{95} = 6.4		A_{95} = 5.8	
		A	21/21	27.3	58.3	64.5	63.5			66.6	173.0
				κ_3 = 47.0		κ_3 = 15.2		α_{95} = 8.4		A_{95} = 6.0	

Bedding, strike/dip with dip direction 90° clockwise from strike. B, A, components B and A. n/N, number of samples or sites used/number of samples or sites measured. D, I, declination and inclination. α_{95}, A_{95}, radius of 95% confidence circle about the mean direction and pole, respectively. For component A, poles and associated A_{95} were calculated from the in-situ directions. κ_3, κ_2, Fisher precision parameters before and after tilt correction. Sites 1-11 are from the Maoping section, and sites 12 -21 are from the Liujiatai section.

= 34.2°/60.7° for the A component at 23% unfolding; yet, this value is not statistically different from that at 0% unfolding (κ = 47.0), suggesting that the A component might be acquired either at the late stage of or after folding. The fact that the paleomagnetic pole (61.0°N/179.0°E, A_{95} = 5.3°) computed from the 23% unfolding falls close to the Jurassic segment of the apparent polar wander path (APWP) for the YB (Enkin et., 1992) and overlaps the pole derived from the A component from Badong in western Hubei after 10% unfolding (Huang and Opdyke, 1996) (Fig. 4) seems to support a late folding origin for the A component. Similar secondary magnetizations have been observed in the Puxi Formation from southeastern Hubei and in the Huangmaqing Formation from Jiangsu (Huang and Opdyke, 1996). Except for the Puxi Fm in which the A component was removed below 550°C and is likely carried by magnetite, the secondary magnetizations from all other formations mentioned above have maximum unblocking temperatures well above 600°C and are thus carried by hematite. The characteristic moderate to steep down inclinations of the secondary magnetizations are also similar to the overprint (S01) observed by Zhao (1987) and Kent et al. (1987) in non-redbed sedimentary rocks with ages ranging from Sinian to lower Triassic from a vast area of eastern YB although the declinations of the (S01) overprint they observed are generally more northerly and dual polarity is occasionally present. Both Zhao (1987) and Kent et al. (1987) have ascribed the overprint to a chemical or diagenetic origin associated with the Yanshanian orogeny in late Mesozoic times. It appears clear that such an overprint is indeed very common in eastern South China. In Fig. 4, the paleomagnetic poles (*PxA*, *ByA* and *BdA*) determined from the A component from different locations seem to follow the late Triassic-Jurassic track of the APWP for the YB, suggesting that the secondary magnetizations might be acquired at slightly different times at different locations. However, the low quality of T3-J segment of the APWP for the YB (Enkin et al., 1992) and the larger uncertainties associated with the pole *PxA* (Huang and Opdyke, 1996) do not allow a firm conclusion to be drawn at present.

The prefolding nature of the B component is further confirmed by McFadden's (1990) fold test result (ξ_1 = 1.241 < 6.919, critic value at the 99% confidence) and a maximum value for κ (29.8) at 98% unfolding. However, the angle between the normal and reverse mean directions is 24.5±16.0°, and McFadden and McElhinny's (1990) reversal test is negative. Apart from insufficient sampling of the reverse polarity, another cause for the failed reversal test might be partial contamination of the ChRM by unremoved A component. Scott and Hotes (1996) have recently developed a method to estimate the magnitude of an unremoved secondary magnetization in a ChRM observed and the deflection caused by the secondary magnetization from the primary direction when the normal and reversed directions are not antiparallel. Applying Scott and Hotes' method to the Liujiatai section, the unremoved A component is estimated to comprise 33% of the ChRM, but it only deflects the mean ChRM direction by about 1.3° (δ). Therefore, the unremoved overprint will not affect the validity of the following discussion.

First, the mean inclination (26.0°) of the B component determined from the Badong Fm in the Jingmen-Dangyang basin is consistent with those observed from Badong in western Hubei (23.0°) and Sangzhi in northwestern Hunan

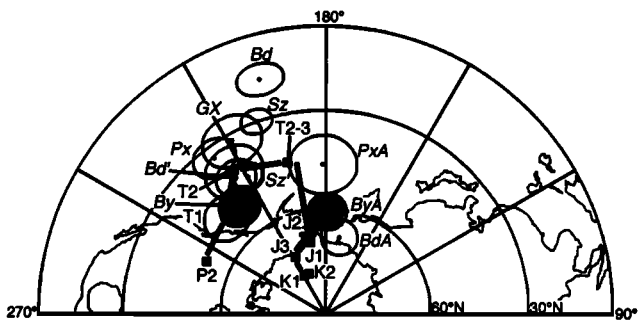


Figure 4. The paleomagnetic poles and associated 95% confidence circles determined from the Middle Triassic rocks from Hubei and northwestern Hunan provinces. *By* = pole computed from the B component of this study (shaded); *ByA* = pole computed from the A component of this study after 23% unfolding (shaded); *Bd* = pole calculated from the B component from Badong, W. Hubei; *Bd'* = pole *Bd* after correction for local rotation; *BdA* = pole calculated from the A component from Badong after 10% unfolding; *Sz* = pole computed from the ChRM from Sangzhi, Hunan; *Sz'* = pole *Sz* after correction for local rotation; *Px* = pole computed from the B component from Puxi, SE. Hubei; *PxA* = pole computed from the A component from Puxi. *GX* is the pole derived by Gilder et al. (1995) from the Middle Triassic from Yufeng, Guangxi province. T2 is the average Middle Triassic pole for the YB computed in this study. T1 is the Early Triassic pole for the YB recalculated by Huang and Opdyke (1996). Other poles determined by Enkin et al. (1992) for the YB are shown with squares: P2-Late Permian; T2-3-Middle-Late Triassic; J1-Early Jurassic; J2-Middle Jurassic; J3-Late Jurassic; K1-Early Cretaceous; K2-Late Cretaceous. Equal-area projection.

(26.8°). These observed inclinations are much steeper than those (17.5±4.7° on average) observed from the Lower Triassic rocks from the YB, supporting the contention that the YB was moving rapidly northward during the interval from Early to Middle Triassic (Huang and Opdyke, 1996). Second, it has been inferred that Sangzhi has been rotated clockwise by 17.2±9.3° with respect to other parts of the YB (Huang and Opdyke, 1996); the mean declination of the B component observed at Yuan'an differs by -25.1±8.6° from that observed at Sangzhi. Thus, Yuan'an is virtually unrotated relative to other parts of the YB (-7.9±12.7°). This is also supported by the fact that the paleomagnetic pole derived from Yuan'an overlaps those from Badong and Sangzhi after correction for the local rotations (Fig. 4). As indicated earlier, the northern flank of the Jingmen-Dangyang basin is involved in the strong deformation of the foreland fold-thrust belt of the QFB, and the sampled Liujiatai section is only 100 km south of the Fangxian-Guangji boundary thrust. If the differential rotations observed in eastern Sichuan (Steiner et al., 1989), western Hubei and northwestern Hunan (Huang and Opdyke, 1996) are due to the collision of the YB with the NCB and subsequent intracontinental thrusting, significant rotations would be expected to take place in the foreland fold-thrust belt. Therefore, the paleomagnetic results from the Jingmen-Dangyang basin seem to disassociate the differential rotations in the above-mentioned area from the collision of the YB with the NCB. Finally, a mean Middle Triassic paleomagnetic pole for the YB (40.9°N/213.1°E, A_{95} = 7.9°, N = 5) has been calculated based upon the poles derived from Guangxi (Gilder et al., 1995), Puxi (Huang and Opdyke, 1996), Yuan'an (this study), and the rotation-corrected poles from Badong (42.9°N/213.6°E) and Sangzhi (44.6°N/209.5°E) (Huang and Opdyke, 1996). This pole together with the recalculated Early Triassic pole (50.5°N/226.1°E, A_{95} = 6.5°, N = 4, see Huang and Opdyke, 1996) and the Late Permian pole constitute a track in the APWP for the YB (Fig. 4) which reinforces the Permo-Triassic swing noted by Enkin et al. (1992) and implies about 9° of northward motion and 24° of counterclockwise rotation of the YB during this time period.

In conclusion, the paleomagnetic results obtained from the Middle Triassic Badong Fm in the Jingmen-Dangyang basin indicates no significant rotation relative to other parts of the YB and thus further delimit the differential rotations to the border area between Sichuan, Hubei, Hunan and Guizhou provinces (Fig. 1). Since one of the sampled sections is in the strongly deformed foreland fold-thrust belt of the QFB, this seems to disassociate these rotations from the collision of the YB with the NCB. Future work should be focused on the border area between Sichuan, Hubei, Hunan and Guizhou, especially eastern Sichuan, to establish the correlation between the rotation and the trend of the fold and look into possible causal relationship of these rotations with the collision of the Huanan Block with the YB (Hsu et al., 1990) and/or the interaction between the Pacific plate and the Asia continent (Sichuan Bureau of Geology and Mineral Resources, 1991).

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